Narin Dhatwalia

Sr. Technical Architect, AI/ML

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Summary

- Hands-on AI/ML architect with 6+ years deploying enterprise AI systems for Fortune 500 clients. At Snowflake, I work directly with customers across healthcare, financial services, telecom, and media, taking LLM applications, agents, and ML models from discovery to production. My fine-tuning idea placed top 3 of 86 internal hackathon teams, and I presented it to the CEO. MS in Machine Learning and Data Science from Northwestern; pursuing a DBA on enterprise GenAI's financial impact.

Experience

Snowflake - Sr. Technical Architect, AI/ML — Evanston IL, USA (Remote) • 06/10/2024 - Present

- Architected a production Cortex AI pipeline leveraging **Claude Sonnet** to analyze clinical data for ~400,000 patients and recommend HCC billing codes with supporting clinical reasoning, projecting ~$34M in new revenue and supporting a **$7.5M contract renewal**. Built an LLM evaluation framework for accuracy, groundedness, and context relevance to monitor outputs.
- Conceived and led a team of five on end-to-end teacher/student distillation: fine-tuned an open-weight model (**Qwen3-8B**) on GPUs from a teacher (**Claude Opus 5.5**). Placed **top 3 of 86 teams** at an internal Snowflake hackathon, and I **presented it to the CEO**. Independent LLM judges score both models on a held-out test set (**lower cost and latency, comparable quality**) so users can decide on Model Registry promotion. Now backed by leadership and Product & Engineering as a native customer feature.
- Designed a Snowflake CoWork solution for a financial services client, replacing manual SQL workflows with a unified Cortex Agent that combines text-to-SQL over a bespoke semantic layer with search over unstructured CRM notes, returning answers in seconds. Instrumented it with evals and AI observability (cost, latency, user feedback, prompt and response traces), enforced governance with Row Access Policies, and worked with Snowflake Product & Engineering on private preview features. Defined the DEV-to-PROD promotion gate for agents with an LLM-as-judge eval suite and pass/fail criteria, replacing subjective spot checks.
- Built ML models across industries: XGBoost router outage prediction with time-series cross-validation for a telecommunications client (**40% less network downtime**; POC contributed to a **$7.2M Snowflake deal**), new-release revenue prediction on a streaming platform for a media client, and transaction categorization for a fintech client covering ~50M payment records weekly.
- Built a text-to-SQL chatbot with Cortex Analyst for a major SaaS company so Customer Success Managers can query account health data in natural language, reducing SQL bottlenecks by 80%.
- Lead workshops and executive readouts for Fortune 500 customers; support pre-sales (discovery, solution design, estimation, SOWs). Spoke to 150+ at an internal summit and won its SOW hackathon track; authored technical content for customers.

TransUnion - Consultant, Data Science & Analytics — Evanston IL, USA (Remote) • 01/02/2023 - 06/07/2024

- Built attribute sets (predictive features of consumer creditworthiness) feeding originations-risk, account management, and marketing propensity models; owned “Soft Inquiry Attributes” end to end (SQL/Spark extraction, R feature calculation) and measured performance lift in credit card marketing models.
- Led a 6-month Data Science Innovation study comparing GPU-accelerated Random Forest against the XGBoost-based originations baseline: significantly lower training time with less overfitting.
- Developed an originations-risk underwriting model using Survival Analysis that predicts the month of potential delinquency, raising AUC from 0.80 to 0.84 and outperforming legacy models.

Decision Point Analytics - Data Scientist — Gurugram, India • 02/13/2020 - 06/04/2021

- Built Bayesian regression Market Mix Models for Kellogg's LATAM to allocate marketing spend across Digital, TV, and Print, incorporating expert input through conjugate priors.
- Used Facebook's Prophet to optimize Coca-Cola Mexico's promotions, decomposing volume lift into forward buying, cannibalization, and true lift, driving a 20% revenue boost the following quarter.

Antuit.ai - Business Analyst — Bengaluru, India • 12/01/2017 - 09/14/2018

- Built automated ETL pipelines across Nielsen, NPD, and IRi data plus Tableau/Power BI dashboards, cutting turnaround by 6 days.

Projects

Text-to-SQL CLI Agent — GitHub portfolio

- Conversational agent for any SQLite database running on an open-weight model served by Fireworks AI, with schema grounding, execution-guided self-repair, and a two-layer guardrail. Graded by executing the SQL and comparing result rows: **9/10** on its evaluation set at 1.15s median latency, versus 1/10 for a naive prompt on the same model. Turning off model reasoning cut median latency 5× and brought projected serving cost to ~$187/month at 30k queries/day, ~13× below GPT-5.4.

Generative AI: Infinite Shakespeare with Transformers — GitHub portfolio

- Built and trained a decoder-only Transformer language model in PyTorch on the complete works of Shakespeare, implementing multi-head self-attention, feed-forward blocks, skip connections, layer normalization, and triangular masking.

Education

Ottawa University, Doctorate in Business Administration — 12/15/2025 - Present

Northwestern University, MS in Machine Learning and Data Science (GPA 3.926) — 09/07/2021 - 12/10/2022

University of Delhi, BS in Mathematics (Honors), 81.46% — 07/01/2014 - 07/01/2017

Skills

Languages and Data: SQL (Snowflake, PostgreSQL), Python, R, Spark (distributed computing), pipeline orchestration

ML and AI: Scikit-learn, XGBoost, PyTorch, TensorFlow, LangChain, LangGraph, RAG, agentic architectures (tool integrations, guardrails), MCP integrations, LLM evaluation (LLM-as-judge), fine-tuning and distillation, open-weight serving (Fireworks AI)

Snowflake (SnowPro Core Certified): Cortex AI (Agents, Search, Analyst), Snowflake CoWork, Snowpark, UDFs/UDTFs, Streamlit, Notebooks, Model Registry, Feature Store, Dynamic Tables, RBAC, Stored Procedures

BI and Databases: Tableau, Power BI, Sigma, Neo4j (Graph Database)

AWS: S3, RDS, ECS, ECR, SageMaker, EC2, VPC, CloudFormation, Route53, AWS CLI